BARCELONA COMPUTATIONAL FOUNDATION

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Pillar: Kolmogorov Theory (KT)

A Lean 4 Formalization of Kolmogorov Theory

Axioms, Theorems, and Project Status

Machine-checking the KT corollaries against an explicit AIT interface

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https://bcom.one · Lean source: https://github.com/giulioruffini/KTAIT ·
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Abstract

We are building `KTAIT`, a Lean 4 + Mathlib formalization whose purpose is narrow and load-bearing: to *machine-check that the Kolmogorov-Theory (KT) corollaries follow logically from an explicit algorithmic-information-theory (AIT) interface*, and that the KT ontology is typed so that structural errors (e.g. comparing a whole pattern to its own part) cannot silently compile. We do *not* re-prove classical AIT: the standard AIT and probability results (the invariance theorem, the coding theorem, symmetry of information, Kraft's inequality, Bayes' rule, the Solomonoff–Levin universal semimeasure, the recursion theorem, Rice's and Chaitin's theorems, and the Vereshchagin–Vitányi structure-function results) are taken as a clearly delimited *axiom layer*. Every *KT* corollary is then *proved* from that layer with no `sorry`, and a toy model witnesses that the axiom layer is jointly satisfiable (so the corollaries are not vacuously true). This working paper states the methodology, fixes the axiom layer, inventories every theorem/lemma across the KT corpus with its current formalization status, and sets a prioritized roadmap.

Keywords: Kolmogorov Theory, algorithmic information theory, Lean 4, Mathlib, formal verification, algorithmic regulator theorem, persistence, self-model.

Classification: MSC 68Q30, 03B35, 68V15 / ACM F.4.1

Contents

1 Introduction and methodology

KT has matured to the point where the dominant risk is no longer in the classical results it invokes, but in whether the *new* KT corollaries are *correctly typed* and *logically entailed* by those results. This project addresses exactly that risk with a proof assistant.[1]

1.1 The one rule

The one rule

Standard AIT (and probability) theorems are allowed to be *axioms*. KT corollaries must be *proved* from those axioms with no `sorry`.

If a KT corollary needs a `sorry`, that is a real gap in the paper. If a classical AIT fact is an axiom, that is honest and expected: we are checking the *dependency structure*, not the AIT foundations.

1.2 Why this is worth doing

The bug class we most want to exclude is *ontological.* For example, writing $\mathrm{I}_K(A : \mathcal{S})$ where $\mathcal{S} \subset A$ is structurally vacuous (a whole versus its own part); the correct object is *temporal*, $\mathrm{I}_K(\mathcal{S}_t : \mathcal{S}_{t+\tau})$ or $K(\mathcal{S}_{t+\tau} \mid \mathcal{S}_t, E, C) \ll K(\mathcal{S}_{t+\tau} \mid C)$. In `KTAIT` the KT roles (*substrate, pattern, self-code, readout, regulator, time*) are *distinct types*; the only bridge to a common carrier is an explicit projection, so the bad version *fails to type-check.* The whole-vs-part error becomes a typed obligation, not a silent mistake.

1.3 Soundness: named hypotheses, not global axioms

A naïve encoding `axiom F : forall (F : AITFrame), P F` asserting a law for *every* frame is *inconsistent* here: one can hand-build a frame violating it (e.g. `slack=0` with mismatched K) and derive `False`, which would make all corollaries vacuously true. We therefore state each AIT fact as a *named hypothesis* (a `Prop` about a frame) and let each corollary consume it. Consequently `#print axioms` on every KT corollary shows only Lean's core axioms (`propext`, `Classical.choice`, `Quot.sound`) — strictly stronger than "rests on named AIT axioms." The toy model (§3) then exhibits a concrete frame satisfying the hypotheses, witnessing joint satisfiability.

1.4 Scope

The current effort is **Level 1** (a typed, axiomatic KT/AIT interface) plus **Level 1.5** (the faithful *probabilistic* ART, including its Bayesian posterior and the coding-theorem bridge). Out of scope for now: a computability-backed K from a concrete universal machine (Level 2), and the differential-geometric (Lie-group / Noether) layer of the world-models paper,[4] which is a separate track.

2 The standard AIT / probability axiom layer

These are the results we *assume.* They are classical and not in doubt; collecting them here fixes precisely what the KT proofs are allowed to lean on. (Per K's directive, starting from these as axioms — and similarly for Bayes — is acceptable and intended.)

Axiom layer A — complexity

Invariance theorem. K is machine-independent up to an additive $O(1)$.
Coding theorem (unconditional). $c_1 2^{-K(x)} \le m(x) \le c_2 2^{-K(x)}$, i.e. $-\log_2 m(x) = K(x) \pm O(1)$.
Coding theorem (conditional). $m(x \mid y) \asymp 2^{-K(x|y)}$.
Kraft–McMillan. $\sum_p 2^{-|p|} \le 1$, hence $\sum_x m(x) \le 1$.
Symmetry of information. $\mathrm{I}_K(x{:}y) = K(x)+K(y)-K(x,y) = K(x)-K(x \mid y^*)+O(\log)$; chain rule $K(x,y) = K(x) + K(y \mid x^*) + O(1)$.
Multiplicity compression. $N_{\le L}(x) \ge 2^r \Rightarrow K(x) \le L - r + O(\log L)$.

Axiom layer B — probability / Bayes

Solomonoff–Levin universal semimeasure. $m(x) = \sum_{U(p)=x} 2^{-|p|}$; conditional $m(R \mid W) \asymp 2^{-K(R|W)}$.
Bayes' rule + deterministic likelihood. prior $P(p) = 2^{-|p|}$, likelihood $P(x \mid p) = \mathbf{1}\{U(p) = x\}$, evidence $P(x) = m(x)$.
Excess-length decay. $\Pr\{|p| \ge K(x) + k \mid x\} \le 2C\, 2^{-k}$.

Axiom layer C — computability / logic

Kleene's second recursion theorem (quines); **Rice's theorem**; **Chaitin's incompleteness** (T proves no $K(x) > n$ for $n > c_T$; K uncomputable); **Vereshchagin–Vitányi**: the Kolmogorov structure function $h_x(\alpha)$ is uncomputable and minimal sufficient statistics are not locatable.

3 Current status of the Lean development

The repository (`https://github.com/giulioruffini/KTAIT`) builds green on Lean 4.31.0 + Mathlib v4.31.0. Modules:

Module	Content	Key results
`Ontology.lean`	six typed KT roles + part-whole guard	`SelfCode` carries `isSub`
`Basic.lean`	`AITFrame`, `IK`, `NMAI`, `condStar`	symmetry as named `Prop`
`Probability.lean`	prefix machine, Bayes posterior, coding thm	Lemma 1 (bridge)
`ART.lean`	`AITProb` (posterior), Theorem 2	probabilistic regulator thm
`Persistence.lean`	`Pers`, `TemporalSelfInfo`	persistence lemmas
`SelfModel.lean`	`DeltaSelf`, Proposition 3	temporal self-model
`ToyModel.lean`	`Toy`, `ToyMachine`, `ToyGap`	satisfiability witnesses
`BadStatements.lean`	guards bite	part-whole + y vs y^*

Proved KT corollaries (no `sorry`; `#print axioms` = Lean core only):

- `AITProb.probabilistic_regulator_theorem` — ART Theorem 2, $\Pr((W,R) \mid x, E) \le C\, 2^{M(W:R)} 2^{-\Delta}$.
- `AITProb.wrapper_bound` — ART Eq. (6), now *derived* from the coding theorem: $\mathrm{post}(e,x) \le (1/c_1) 2^{K(x)-K(e)}$.
- `AITProb.probabilistic_regulator_theorem_sharp` — ART Theorem 2 *sharp form* retaining the regulator cost: post $\le C\, 2^{M(W:R)} 2^{-\Delta} 2^{-K(R)}$ (no extra hypothesis; the headline drops $2^{-K(R)} \le 1$, `..._of_sharp`).
- `AITProb.theorem1_posterior_tilt` — ART Theorem 1, two-sided posterior tilt.

- `AITProb.theorem3_onoff_evidence` — ART Theorem 3, ON/OFF evidence ratio $\asymp 2^{\Delta}$ (log-free form).
- `CoarseGraining.theoremB` / `corollaryB` — WP0193 Theorem B / Corollary B: regulatory coarse-graining is uncomputable, by reduction to Vereshchagin–Vitányi.
- `quine_floor`, `self_prediction_dichotomy`, `chaitin_blocks_minimality` — WP0192 Principle 1 / WP0162 Proposition 4: the three independent obstructions to a complete, exact, certifiable self-model (the middle one is an axiom-free diagonalization).
- `contrast_posterior_ranks_by_complexity` — WP0162 Q3 (Eq. 26): the contrast-fiber posterior ranks by joint simplicity, with no $2^{-\Delta}$ tilt.
- `OrbitLabel.genEnergy_conserved` — WP0162 App. C: generalized energy as a conserved orbit label (axiom-free).
- `boundary_sufficient` — WP0162 App. E: the thin boundary as a compressed sufficient statistic.
- `low_complexity_shrinkage` — ART Theorem A4; `CoarseGraining.theoremA`, `CoarseGraining.existence` — WP0193 Theorem A and Proposition 1.
- `Noether.trajLabel_conserved`, `Noether.conserved_comp_symm` — the abstract core of generalized Noether (Structured Dynamics paper): the trajectory label is conserved by the flow, and symmetries commuting with the flow act on conserved quantities. (Continuous-time analogue of the orbit-label theorem.)
- `Homogeneous.homogeneousSpace` — a transitive symmetry action makes the state space a homogeneous space, $X \simeq G/H_x$ (Theorem A4's algebraic core; the manifold/dimension layer remains future work).
- `persistence_conservation` / `conservation_tradeoff` — WP0162 Proposition 2: the ledger $K(O_W) = \mathrm{I}_K(O_W{:}R) + K(O_W \mid R^*)$ (from symmetry of information), and at fixed $K(O_W)$, maximizing I_K = minimizing the conditional residual.
- `regulator_selection` — WP0162 Proposition 1: $\arg\min_R K(R \mid W) = \arg\max_R [M(W{:}R) - K(R)]$.
- `AITProb.probabilistic_regulator_theorem_conditional` — ART sharp form in the $2^{-K(R|W)}$ reading (posterior favors regulators simple given the world).
- `PrefixMachine.lemma1_posterior_bounds` — the Bayes$\leftrightarrow$Kolmogorov bridge, $\frac{1}{c_2} 2^{K(x)-|p|} \le P(p \mid x) \le \frac{1}{c_1} 2^{K(x)-|p|}$.
- `self_regulation_temporal_model` — Proposition 3, $K(S_{t+\tau} \mid S_t, E, C) \le K(S_{t+\tau} \mid C) - \Delta_{\text{self}} + 2\,\text{slack}$.
- `pers_eq_nmai`, `persistent_pos` — persistence as temporal self-information.

Satisfiability witnesses: `toy_symmetry` (symmetry of information holds for `Toy`), `toyMachine_coding` (coding theorem holds for `ToyMachine`), `toy_self_model_fires` (the self-model corollary fires with a genuine gap $\Delta_{\text{self}} = 3$). **Guards:** `#check_failure (peers A S)` (part-whole), and `ystar_form_holds` vs `rawy_form_fails` (symmetry must use y^*, not y).

4 Theorem inventory and formalization status

Every numbered/named formal statement across the KT corpus, classified **[STD]** (standard AIT/probability/logic — axiom candidate) or **[KT]** (Kolmogorov-Theory-specific — proof target), with status. Sources: ART paper,[2] Pattern Persistence (WP0162),[3] Agent Know Thyself (WP0192), Regulatory Coarse-Graining (WP0193), world-models paper,[4] and the foundational AIT-of-consciousness paper.[5]

ID (paper)	**Statement (abbreviated)**	**Class**	**Status**
ART paper — The Algorithmic Regulator (Entropy 2026, 28, 257)			
Lemma 1	$P(p \mid x) = 2^{-\lvert p\rvert}/m(x)$; sandwiched by coding thm	STD	**Proved** (`lemma1_...`)
Theorem 2	$\Pr((W,R) \mid x, E) \le C\,2^{M(W:R)}2^{-\Delta}$	KT	**Proved** (`prob_reg_thm`)
Theorem 2 (sharp)	retains regulator cost: $\le C\,2^{M}2^{-\Delta}2^{-K(R)}$	KT	**Proved** (`.._sharp`)
Lemma 2	OFF run lower-bounds world: $K(O_{W,\varnothing}) \le K(W) + c_0$	KT	**Axiom** (hyp. `hoff`)
Eq. (6)	wrapper bound post $\le (1/c_1)2^{K(x)-K(W,R)}$	KT	**Proved** (`wrapper_bound`)
Theorem 1	coupled-pair posterior tilt $\in [\cdot]2^{K(x)-K(W)-K(R)+M}$	KT	**Proved** (`theorem1_...`)
Theorem 3	$m(O_{W,R})/m(O_{W,\varnothing}) \asymp 2^{\Delta}$ (as-if objective)	KT	**Proved** (`theorem3_...`)
Theorem A4	low-complexity output $\Rightarrow$ strict tiny shrinkage of $K(W \mid R)$	KT	**Proved** (`low_complexity_shrinkage`)
Defs 1–2	internal model ($M(W{:}R) > 0$); good regulator ($\Delta > 0$)	KT	n/a (encoded)
Thms A1–A3, Lem A1	coding thm (uncond/cond), Kraft counting, excess-length decay	STD	**Axiom**
Pattern Persistence — WP0162			
Proposition 3	self-regulation $\Rightarrow$ temporal self-model (Eq. 44)	KT	**Proved** (`self_reg_...`)
Proposition 1	regulator selection $\arg\min K(R \mid W) = \arg\max[M - K(R)]$	KT	**Proved** (`regulator_selection`)
Proposition 2	persistence conservation ledger $K(O_W) = \mathrm{I}_K + K(O_W \mid R^*) \pm O(\log)$	KT	**Proved** (`persistence_...`)
Proposition 4	self-model incompleteness (3 obstructions)	KT	**Proved** (`quine_floor`,...)
Orbit-label thm	conserved orbit label = "generalized energy"	KT	**Proved** (`genEnergy_conserved`)
Agent Know Thyself — WP0192			
Proposition 1	temporal self-model (= WP0162 Prop. 3)	KT	**Proved** (same)
Principle 1	self-model incompleteness (= WP0162 Prop. 4)	KT	**Proved** (Phase 4)
Conjecture	acquired self-model prior	KT	n/a (conjecture)
Regulatory Coarse-Graining — WP0193			
Theorem B	targeted sufficient projection uncomputable	STD	**Proved** (`theoremB`)
Corollary B	regulatory coarse-graining uncomputable	KT	**Proved** (`corollaryB`)

ID (paper)	Statement (abbreviated)	Class	Status
Lemma 1	uncomputability of $h_{x\to y}$	STD	**Axiom** (via V–V)
Proposition 1	existence of a bounded regulation-sufficient projection	KT	**Proved** (`existence`)
Theorem A	selection uncomputability	KT	**Proved** (`theoremA`)
Cor. C1–C3	blanket uncomputability/fuzziness; acquisition; discovery	KT	n/a (interp.)
World models / Lie groups — Entropy 2025 (separate geometry track)			
Thms A1–A5	Lie transitivity, homogeneous space, moduli stack	STD	n/a (out of scope v1)
Noether (gen.)	continuous symmetry ⇒ conserved quantity (abstract flow core)	KT	**Proved** (`trajLabel_conserved`)
Homogeneous space (transitive action ⇒ G/H)	—	STD	**Proved** (`homogeneousSpace`)
Lie pseudogroups / moduli stacks / Mostow / dim formula	—	geom	**TODO** (not in Mathlib)
Foundational — nix019 / K-input			
Hyp. 1–2, Cons. 1–2	simplicity selection; structured experience; high MAI	KT	n/a (informal)

Legend. **Proved**: machine-checked in Lean, no `sorry`. **Axiom**: assumed (standard result, or a named hypothesis consumed by a corollary). **TODO**: a KT proof target not yet formalized. n/a: definition, conjecture, interpretive remark, or informal claim (not a proof target in the current scope).

5 Roadmap

Priorities are ordered to maximize KT-correctness assurance per unit effort, exploiting the fact that the AIT/Bayes layer (§2) may be assumed.

Phase 1 — close the ART chain. DONE.
The wrapper bound Eq. (6) is now a *derived* theorem (`wrapper_bound`) from the coding theorem (`CodingLB`); ART **Theorem 1** (`theorem1_posterior_tilt`) and **Theorem 3** (`theorem3_onoff_evidence`, log-free form) are proved; `probabilistic_regulator_theorem` now consumes `wrapper_bound` and rests on the coding theorem + Lemma 2. All four `#print axioms` = Lean core only.

Phase 2 — persistence conservation (WP0162 Prop. 2). DONE.
Ledger $K(O_W) = \mathrm{I}_K(O_W{:}R) + K(O_W \mid R^*)$ (`persistence_conservation`, from symmetry of information) and the trade-off "max I_K = min conditional residual" (`conservation_tradeoff`).

Phase 3 — regulator selection (WP0162 Prop. 1). DONE.
$\arg\min_R K(R \mid W) = \arg\max_R[M(W{:}R) - K(R)]$ proved (`regulator_selection`) from the chain rule; the ART sharp form's $2^{-K(R|W)}$ reading is `probabilistic_regulator_theorem_conditional`.

Phase 4 — self-model incompleteness. DONE.
The three independent obstructions: the quine floor (`quine_floor`, $|Q_A| \geq K(A)$), the self-prediction dichotomy (`self_prediction_dichotomy`, an axiom-free diagonalization), and the Chaitin ceiling on minimality (`chaitin_blocks_minimality`).

Phase 5 — coarse-graining uncomputability (WP0193). DONE.
Theorem B and Corollary B (regulatory coarse-graining uncomputable) proved by an abstract reduction to Vereshchagin–Vitányi (`theoremB`, `corollaryB`).

All five roadmap phases are complete. The WP0162 / WP0192 / WP0193 KT corollaries and the full probabilistic ART layer are machine-checked, `sorry`-free, resting only on Lean core plus the named AIT axiom layer of §2.

Later — geometry track.
The Lie-group/Noether layer of the world-models paper is a distinct development (differential geometry in Mathlib), deferred until the AIT/KT core is complete.

Each phase keeps the invariant: KT results `sorry`-free; `#print axioms` = Lean core + named hypotheses; new hypotheses discharged by an extended toy model.

6 Conclusion

We have a green, machine-checked Level-1/1.5 core: the probabilistic ART (Theorem 2), the Bayes↔Kolmogorov bridge (Lemma 1), the temporal self-model (Proposition 3), and persistence, all proved from an explicit, satisfiable AIT/Bayes axiom layer, with the KT ontology typed so the whole-vs-part error cannot compile. This document is the living status tracker; the roadmap above closes the remaining KT corollaries paper-by-paper while keeping the soundness invariants intact.

References

[1] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, CADE 2021; The mathlib Community, *The Lean Mathematical Library*, CPP 2020.

[2] G. Ruffini et al., *An Algorithmic-Information-Theoretic Regulator Theorem*, Entropy **28**(3):257, 2026.

[3] G. Ruffini, *Pattern Persistence* (BCOM WP0162), 2026.

[4] G. Ruffini et al., *World models, Lie groups, and symmetry in Kolmogorov Theory*, Entropy **27**(1):90, 2025.

[5] G. Ruffini, *An algorithmic information theory of consciousness*, Neuroscience of Consciousness **2017**(1):nix019.

[6] M. Li and P. Vitányi, *An Introduction to Kolmogorov Complexity and Its Applications*, 4th ed., Springer, 2019.